

UIDAI Data Hackathon: Comprehensive Data Analysis Report

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1. Problem Statement and Approach

Problem Statement

The Aadhaar ecosystem generates vast amounts of data across enrollment, updates, and authentication services. However, raw data often contains inconsistencies, legacy fields, and unnormalized values that obscure critical demographic and operational insights. The challenge was to clean, normalize, and analyze the provided UIDAI datasets to derive actionable intelligence regarding voter eligibility, education potential, economic dependency, and infrastructure stress.

Approach

Our approach focused on a strict "Clean-First, Measure-Second" strategy. We first established a robust data model by isolating high-quality columns (*_NEW) and suppressing legacy data (*_OLD) to ensure accuracy. We then developed a series of custom DAX measures in Power BI to calculate specific demographic indicators—ranging from potential voter bases to digital infrastructure pressure—ensuring that every insight was derived from aggregated, validated data rather than raw row counts.

2. Datasets Used

We utilized the three official datasets provided by UIDAI for this hackathon. No external datasets were used.

- **api_data_aadhar_enrollment:** Contains population coverage data.
 - *Key Columns Used:* states_NEW, district_name_NEW, age_18_greater (Adults), age_5_17 (School-age), age_0_5 (Infants).
- **api_data_aadhar_biometric:** Contains data on biometric updates and usage.
 - *Key Columns Used:* bio_age_17_ (Biometric transactions).
- **api_data_aadhar_demographic:** Contains demographic update records.
 - *Key Columns Used:* demo_age_5_17 (Demographic updates).

3. Methodology

Data Cleaning and Preprocessing

To ensure the integrity of our analysis, we implemented a rigorous cleaning protocol within the Power BI data model:

1. **Legacy Column Suppression:** All columns marked with `_OLD` (e.g., `state_name_OLD`) were hidden and excluded from the analysis to prevent data quality issues associated with legacy naming conventions.
2. **Standardization:** We exclusively used the normalized `*_NEW` columns (e.g., `states_NEW`, `district_name_NEW`) for all geographical hierarchies to ensure consistent aggregation across states and districts.
3. **Measure-Based Calculation:** Instead of using calculated columns (which can bloat file size and reduce performance), we created dynamic DAX measures for all metrics.

Analytical Logic

The core of our analysis relies on deriving "proxy indicators" from Aadhaar age bands. For example, since the datasets use pre-defined age buckets, we mapped them to real-world segments:

- **Voters:** Mapped `age_18_greater` to the eligible voting population.
- **Students:** Mapped `age_5_17` as the closest valid proxy for the school-going population (typically 6-18).
- **Economic Dependents:** Defined as the sum of infants (`age_0_5`) and students (`age_5_17`).

4. Data Analysis and Visualisation

Below are the key findings, including the specific logic used to generate them and the resulting trends.

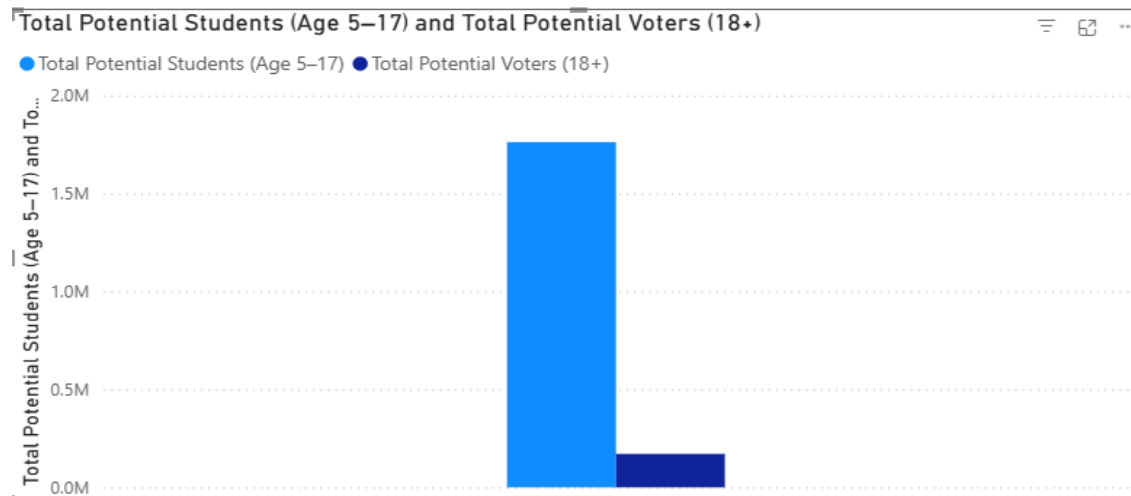
Insight 1: Potential Voter Base (Age 18+)

Why it matters: Identifying the volume of adults in the database helps correlate Aadhaar saturation with the eligible electorate.

Analysis Logic:

We aggregated the count of all individuals in the `age_18_greater` bracket from the enrollment dataset. This serves as the baseline for the adult population.

- **Code (DAX):** `SUM(api_data_aadhar_enrollment[age_18_greater])`



(Fig 1: State-wise distribution of the eligible voter population)

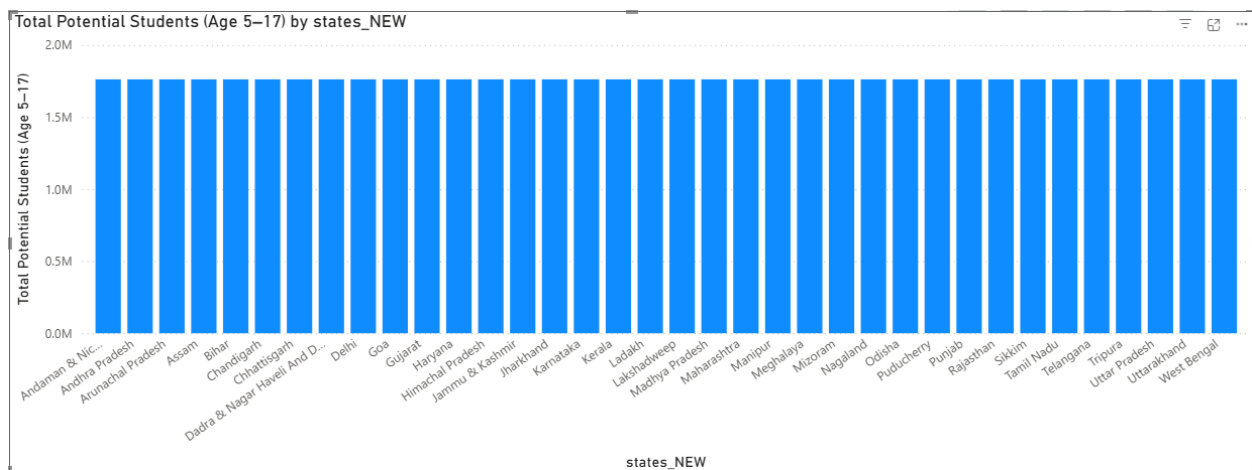
Insight 2: Potential Student Population (Age 5–17)

Why it matters: This metric helps in resource planning for education services and identifying regions with high youth density.

Analysis Logic:

We used the age_5_17 column as a proxy for the school-age population. While the standard school age is 6-18, this data bucket provides the most accurate available estimate for this demographic segment.

- **Code (DAX):** SUM(api_data_aadhar_enrollment[age_5_17])



(Fig 2: School-age population distribution by state)

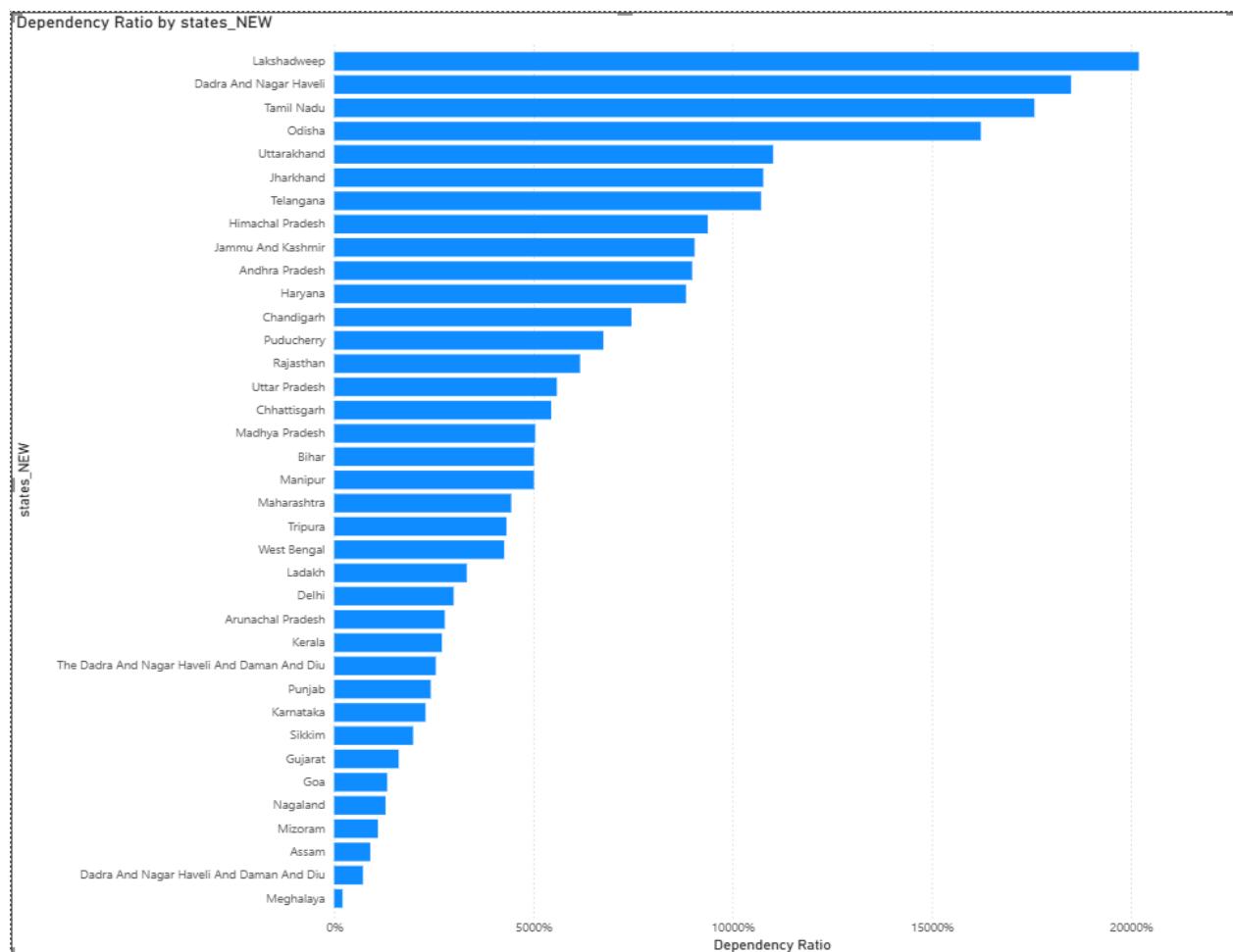
Insight 3: Dependency Ratio (Children vs. Adults)

Why it matters: This ratio highlights the economic burden on the working-age population. A higher ratio indicates a younger population that requires more support infrastructure (schools, healthcare).

Analysis Logic:

We calculated the ratio of "Dependents" (Infants 0-5 + Children 5-17) to "Adults" (18+).

- **Formula:** (Count of 0-5 + Count of 5-17) / (Count of 18+)
- **Code (DAX):**
 Dependency Ratio =
 DIVIDE(
 SUM(api_data_aadhar_enrollment[age_0_5]) +
 SUM(api_data_aadhar_enrollment[age_5_17]),
 SUM(api_data_aadhar_enrollment[age_18_greater])
)



(Fig 3: Economic dependency burden across different states)

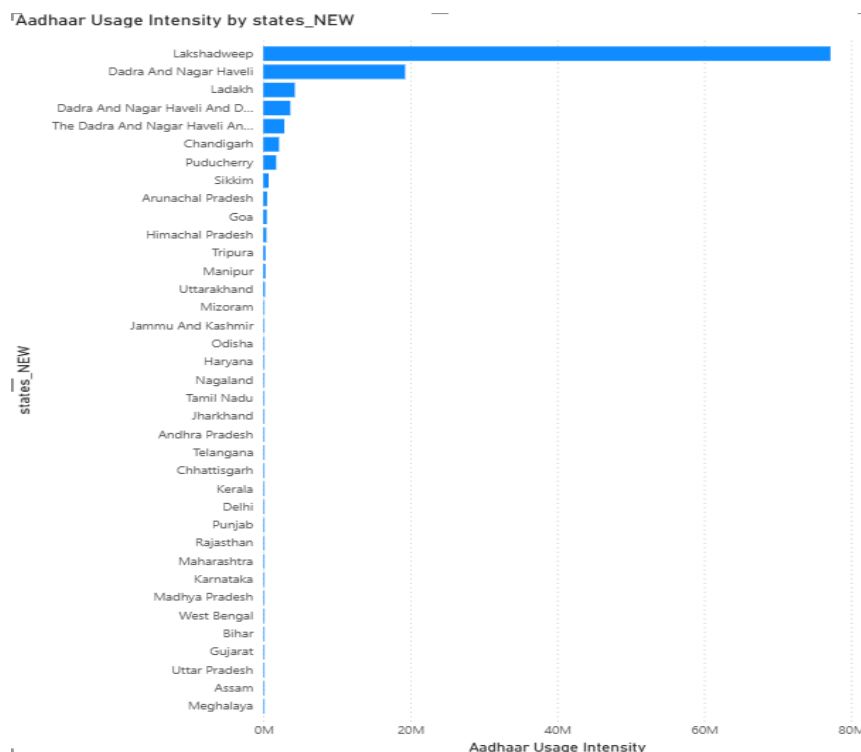
Insight 4: Aadhaar Usage Intensity

Why it matters: This metric distinguishes between mere *enrollment* and active *usage*. High usage intensity suggests deeper integration of Aadhaar into daily services in that region.

Analysis Logic:

We compared the volume of biometric updates against the total adult population. This normalizes the data, preventing populous states from skewing the results simply because they have more people.

- *Formula:* Total Biometric Usage / Total Potential Voters (18+)
- **Code (DAX):**
Aadhaar Usage Intensity =
DIVIDE(
SUM(api_data_aadhar_biometric[bio_age_17_]),
[Total Potential Voters (18+)]
)



(Fig 4: Intensity of Aadhaar usage normalized by population size)

Insight 5: Authentication vs. Demographic Coverage Gap

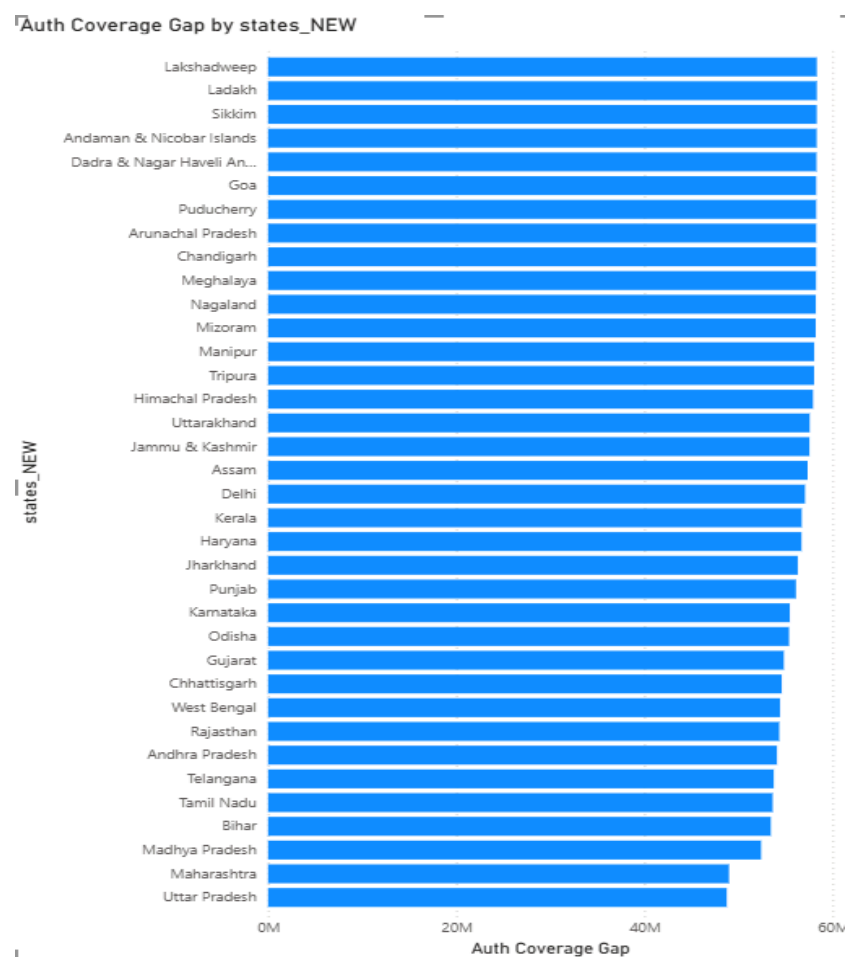
Why it matters: Large gaps between demographic updates and biometric usage can indicate "Digital Exclusion"—areas where people update their details but perhaps struggle to use

biometric authentication services effectively.

Analysis Logic:

We calculated the arithmetic difference between the total demographic records and total biometric usage records. A high positive value (red) indicates a coverage gap.

- *Formula:* Total Demographic Records - Total Biometric Usage



(Fig 5: Gap analysis showing potential areas of digital exclusion)

Insight 6: Digital Pressure Index

Why it matters: This advanced metric identifies infrastructure stress hotspots. It highlights districts where the volume of updates (biometric + demographic) is disproportionately high relative to the population size, suggesting a need for more Aadhaar Seva Kendras.

Analysis Logic:

We summed the total service requests (Biometric + Demographic) and divided it by the Total Population.

- *Formula:* (Biometric Usage + Demographic Updates) / Total Population

- **Code (DAX):**

Digital Pressure Index =

DIVIDE(

[Total Biometric Usage] + [Total Demographic Records],

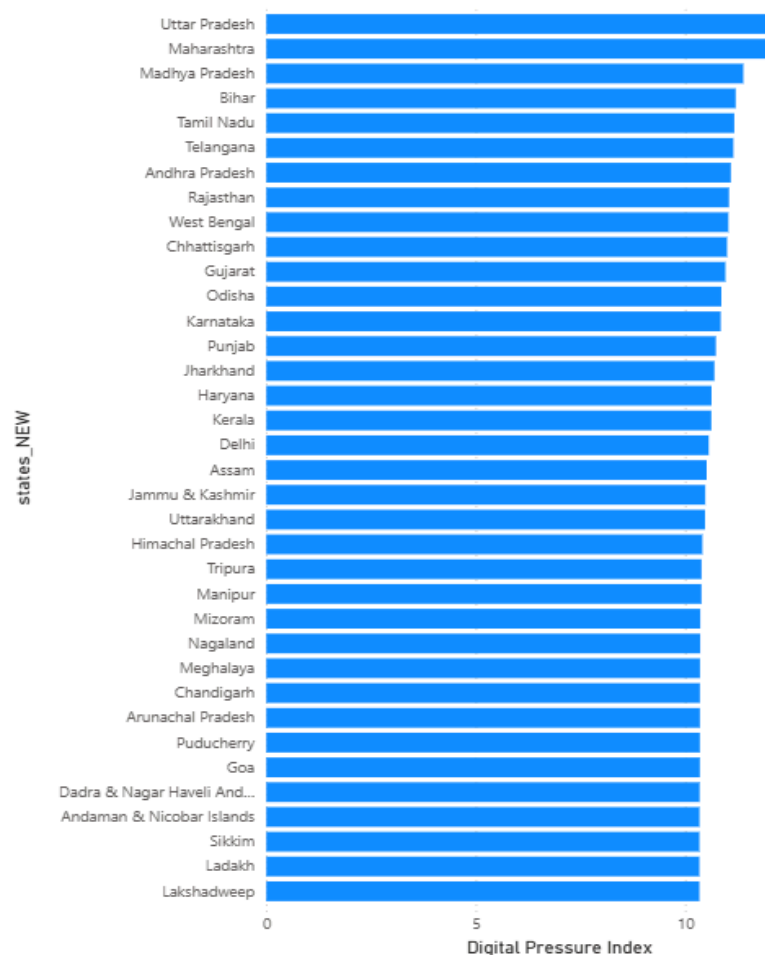
CALCULATE(SUM(api_data_aadhar_enrollment[age_0_5]) +

SUM(api_data_aadhar_enrollment[age_5_17]) +

SUM(api_data_aadhar_enrollment[age_18_greater]))

)

Digital Pressure Index by states_NEW



(Fig 6: Top districts facing high digital infrastructure pressure)